



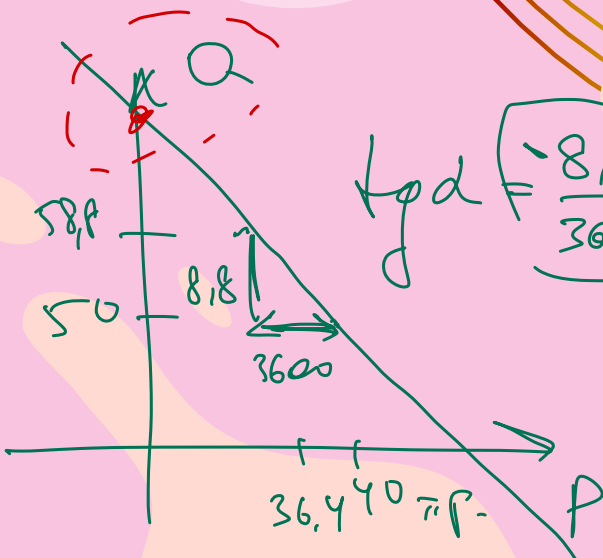
$$50 = y = -\frac{8,8}{3600} 4000 + b$$

$$\left(50 + \frac{8,8}{3600} 4000 = b \right)$$

$$k = -\frac{8,8}{3600}$$

$$(x_1 : y_1) \quad (x_2 : y_2)$$

$$\begin{cases} y = kx + b \end{cases} \rightarrow (k, b)$$



$$k = \frac{-8,8}{3600}$$

14

2

$$\Delta Q\% = -1,75\%$$

$$\Delta Q\%_2 = 0,7\%$$

$$\varepsilon = -0,8$$

$$40000p = P_1$$

$$36400p = P_2$$

$$Q_1 = 50$$

$$Q_2 = 58,8$$

$$\Delta Q\% = 17,6\%$$

$$\Delta Q\% = \varepsilon \Delta P = -0,8 \cdot (-8\%) = 17,6\%$$

$$\Delta TR\% \approx \Delta Q\% + \Delta P\% = 17,6\% - 8\% = 9,6\%$$

$$T_1 = Q_1 \cdot P_1$$

$\begin{matrix} \text{"} & \text{"} \\ 50 & \cdot & 40000 \end{matrix}$

$$T_2 = Q_2 \cdot P_2$$

$$100\% Q_1 = 50$$

$$117,6\% Q_2 = x$$

$$Q_2 = x = \frac{117,6 \cdot 50}{100}$$

$$P_2 = 0,92 \cdot 40000$$

$$T_2 - T_1 = nTR$$

$$x = \Delta P \% \quad \uparrow \quad \text{ATR} \approx \underline{\underline{1\%}}$$

на сколько процентов?

$$\Delta Q \% = \varepsilon \cdot X = -2,2x$$

$$\text{ATR \%} = \underset{-2,2x}{\Delta Q \%} + \underset{x}{\Delta P \%} = 1\%$$

$$-2,2x + x = 1\%$$

$$-1,2x = 1\%$$

$$\rightarrow x = \frac{1}{-1,2}$$

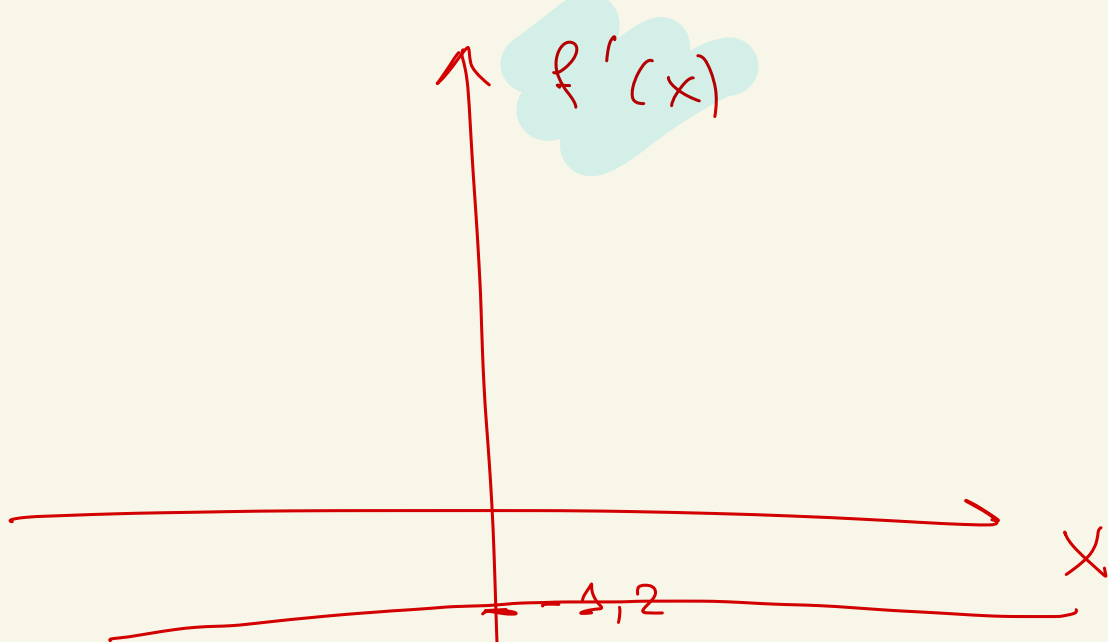
$$\varepsilon = -2,2$$

$$\Delta \text{TR \%}$$

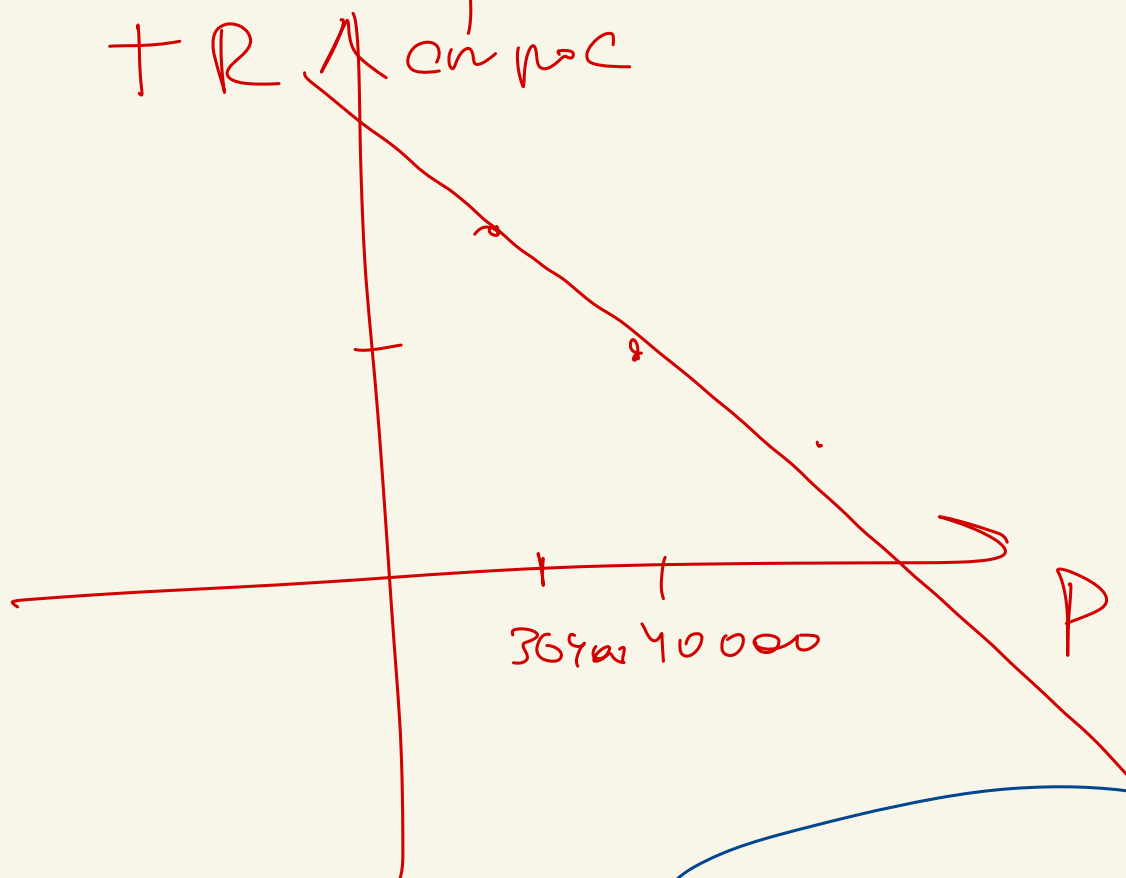
$$\Delta P \cdot \varepsilon = \Delta Q$$

$$\Delta \text{TR} = X - 2,2X = -1,2X$$

$$(\Delta \text{TR})' = -1,2 \Rightarrow \underline{\underline{\text{ATR монет. укл}}}$$



+ R на по C



$$Q = k_p + b$$

$$Q = -P + 20$$

$$\text{TR} = Q \cdot P = (kP + b)P = kP^2 + bP$$

$$(\text{TR})' = 2kP + b = 0$$

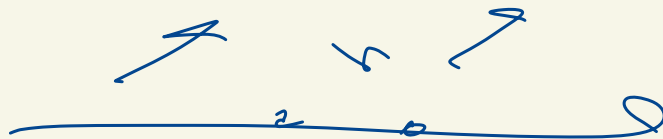
$$P = -\frac{b}{2k} \leftarrow \text{max}_{\text{ген}}$$

$$Q = 2^3 - P^2 + 2P + 1$$

max (TR)

$$\text{TR} = P \cdot Q = P^4 - P^3 + 2P^2 + P$$

$$\text{TR}' = 4P^3 - 3P^2 + 4P + 1 = 0$$



$$e^{2x} - 6e^x = 0$$

$$e^x = 6$$

$$e^x = 0 \Rightarrow \emptyset$$

$$e^x = 6$$

$$\ln e^x = \ln 6$$

$$x = \ln 6$$



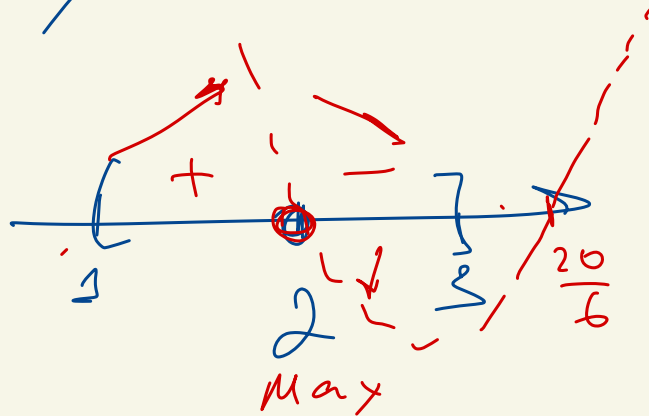
$$y = e^{2\ln 6} - 6e^{\ln 6} + 3 =$$

$$6^2 - 6 \cdot 6 + 3 = 3$$

$$\frac{20}{6}$$

$$2$$

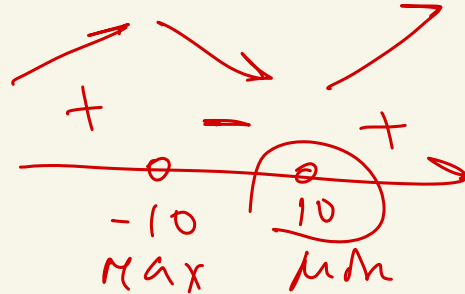
$$3x^2 - 16x + 20$$



$$y' = \frac{-100}{x^2} + 1$$

$$\left(\frac{100}{x}\right)' = \frac{0 \cdot x - 1 \cdot 100}{x^2} = -\frac{100}{x^2}$$

$$y' = \frac{-100 + x^2}{x^2} = \frac{x^2 - 100}{x^2}$$



$$y = \sqrt{1 - 4x - x^2}$$

$$y' = \frac{(-2x - 4)}{2\sqrt{1 - 4x - x^2}} = 0$$

$$-2x - 4 = 0$$

$$2x = -4$$

$$x = -2$$

$$\frac{a^2}{a^{1/2}}$$

$$y(-2) = \sqrt{1 + 8 - 4} = \sqrt{5}$$



$$y' = -5 \sin x - 6$$

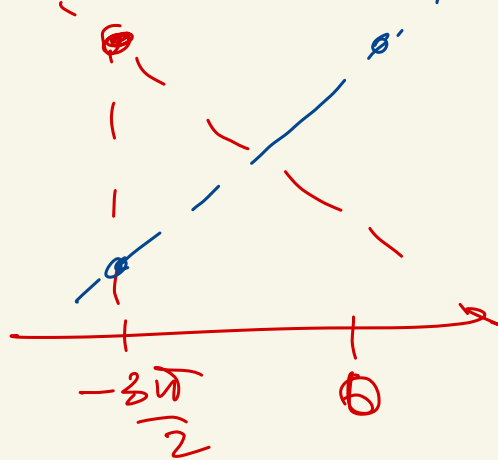
$$\sin x = -\frac{6}{5}$$

корней нет

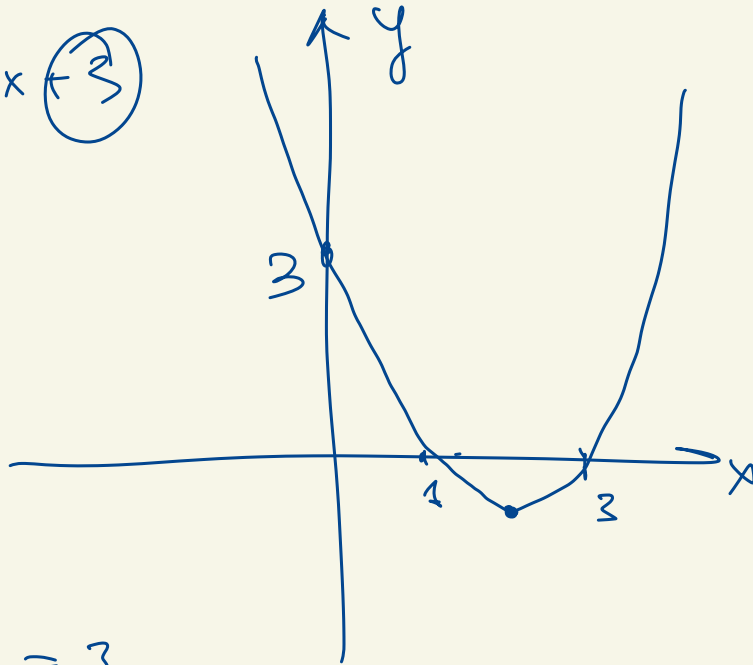
всегда +

или

всегда -



$$y = x^2 - 4x + 3$$



$$x_0 = 3$$

$$y = f'(x_0)(x - x_0) + f(x_0)$$

$$f'(x_0) = 2x - 4 = 2$$

$$f(x_0) = 0$$

$$y = 2(x-3)$$

$$-1 + 3 = ?$$

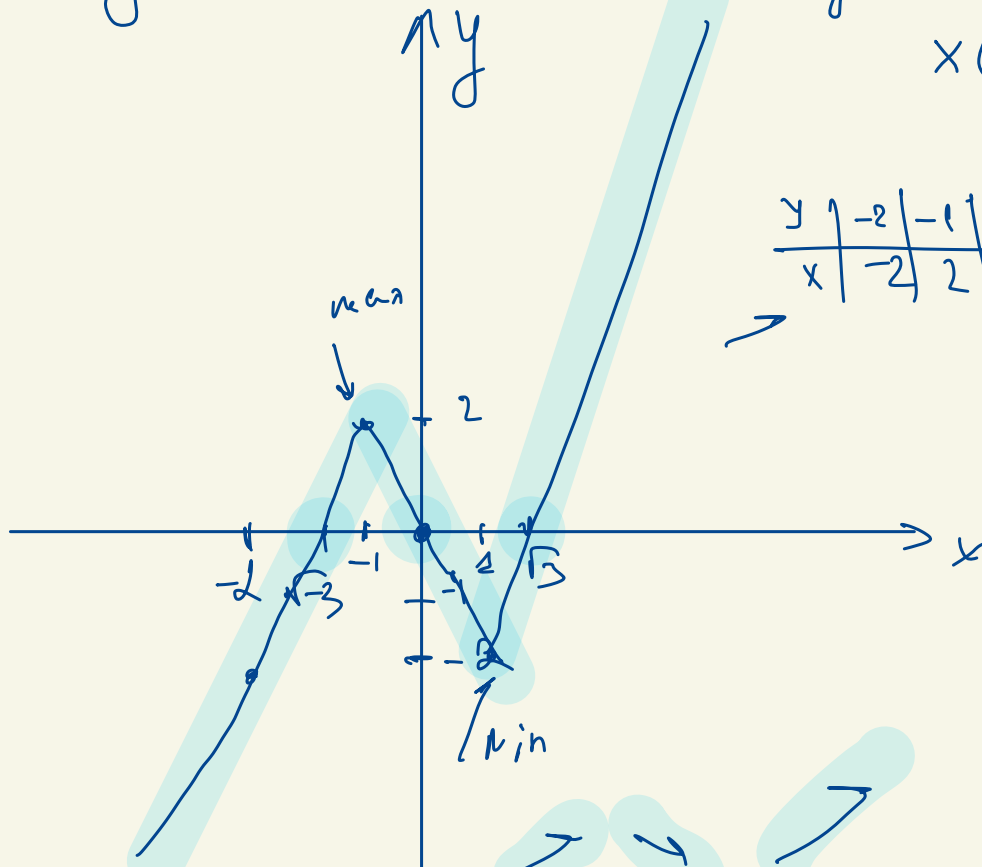
$$y = x^3 - 3x$$

$$x(x^2 - 3) = 0$$

$$x = 0$$

$$x = \pm\sqrt{3}$$

y	-2	-1	0	1
x	-2	2	0	-2



$$y' \Rightarrow \pm 1$$

